Seat No: .... **Enrollment No:** ....

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech/Int. B.Tech, Winter 2025-26 Examination

Semester: VII/ XI **Date: 08-12-2025**
Subject Code: 203105310 **Time: 2.00 pm to 4.30 pm**
Subject Name: Information and Network Security **Total Marks: 60**

Instructions:
1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	**Marks**	**CO**	**BT**
	A. a) Define access control. b) List any two availability services. c) State the goal of data origin authentication.	06	CO1	BT1
	B. a) Explain attacker's advantage in chosen-plaintext. b) Differentiate chosen-plaintext from known-plaintext. c) Summarize why stream ciphers vulnerable.	06	CO1	BT2
Q.2	A. Analyze effective key size of 2-key Triple DES.	04	CO3	BT4
	B. Implement CTR encryption using counters [0001, 0010, 0011]	05	CO1	BT3
	OR			
	B. Apply CFB to encrypt binary stream sequentially.	05	CO1	BT3
Q.3	A. Apply initial permutation to binary input "10110011"	04	CO2	BT3
	B. Analyze differential trail in simplified Feistel cipher	05	CO1	BT4
	OR			
	B. Show keying option 3 in Triple DES.	05	CO1	BT3

SECTION - B

Q.1	Answer the following questions.	**Marks**	**CO**	**BT**
	A. a) Define GF(p). b) What does 'p' represent in GF(p)? c) Give an example of GF(5).	06	CO1	BT1
	B. a) Explain confidentiality in PKC. b) Explain authentication in PKC. c) Explain non-repudiation in PKC.	06	CO1	BT2
Q.2	A. Analyze CBC-MAC failure on variable-length messages	04	CO2	BT4
	B. Show trojan's effect on user authentication process.	05	CO4	BT3
	OR			
	B. Generate digital signature with given RSA key pair	05	CO3	BT3
Q.3	A. Analyze how firewalls prevent intruder access to key stores.	04	CO4	BT4
	B. Analyze consequences of MITM on data integrity	05	CO1	BT3

	OR			
	B. Analyze CBC-MAC security under variable message lengths	05	CO4	BT4